Serologic Markers of Neurologic Diseases: The Control Groups

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Increasing the diagnostic armamentarium for stroke and other neurologic conditions is of interest to the neurologic and medical communities. Thus, the recent publication by Jung and colleagues, who studied circulating endothelial microparticles (EMPs), may advance the field of serologic stroke diagnosis by serology.[1] As the relevance of various biomarkers becomes elucidated, including EMPs, it will be important to identify key control groups in these studies. Because EMPs have been identified in a variety of neurologic and non-neurologic disorders,[2] the biological and clinical importance of these biomarkers may be unclear when evaluating them only in the context of one neurologic disease, such as stroke. Thus, alongside the EMP levels in stroke, it will be valuable to determine EMP levels in patients with other vascular, inflammatory, infectious, neoplastic, and degenerative neurologic conditions.

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Reply

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We would like to thank Dr Cohen for his interest in our paper. Serologic testing is the most useful tool for sensitive screening of large populations at risk for stroke and to determine an individual's baseline for a diagnostic and therapeutic approach. Endothelial microparticles (EMPs) in the circulation differentially increase in populations with different phenotypes, and easy detection in daily practice would make EMPs potentially useful candidates for biomarkers of cerebrovascular disease (CVD).[1] CVD is heterogeneous with respect to clinical manifestations and pathophysiology. The EMP profile is instrumental for risk stratification of CVD, and its numeric scale allows determination of the best cutoff values of EMPs for CVD. In our previous study,[1] the combined model with $CD62E^+$ EMP $\geq 100/\mu l$ and $CD62E^+$ to $CD31^+AV^+$ EMP ratio ≥ 2.0 provided a sensitivity of 73% and a specificity of 82% for extracranial artery stenosis. The strength of this technology in assay accuracy, precision, and reproducibility can be beneficial for an improved diagnosis of CVD, as well as for monitoring of disease activity and treatment response. Future clinical application with improved diagnostic power will be achieved by large-scale longitudinal trials using careful specimen collection and standardized, quality-controlled data collection. Another point to consider is that EMPs do not reflect the tissue of origin. With other vascular or tissue injury, EMPs can be released into the circulation in a nonspecific manner.[2] Although other well-defined factors with endothelial pathology were excluded in our study,[1] the presymptomatic biologic changes might have influenced blood endothelial component levels. Furthermore, the pathogenesis of neurodegeneration is thought to involve neurovascular dysfunctions, including attenuated capillary density, endothelial dysfunction, impaired vasomotor activity, and decreased cerebral perfusion.[3,4] Because our study was limited to patients at risk for CVD,[1] we do not know the extent of the influence of other disease conditions. EMP analysis, therefore, should be re-emphasized by applying its clinical use to the prediction or diagnosis of various vascular and neurologic diseases.

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Association of Pyridoxal Kinase and Parkinson Disease

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The identification of novel genetic factors that influence susceptibility to Parkinson disease (PD) is the driving force behind the generation of in vitro and in vivo model systems and thus the development of targeted therapeutics. To date, the genes implicated in PD have been identified through classical linkage studies in extended multi-incident kindreds.[1] However, the identification and collection of such large PD families is rare, and often takes many years[2]; alternative approaches including association, gene expression, and protein studies have been applied to address the problem.

We read with interest the recent study by Elstner et al, which compared whole genome expression patterns in isolated neurons from the substantia nigra of patients with PD and healthy control subjects.[3] Four genes were shown to have different expression levels between the 2 groups (*MTND2*, *SRGAP3*, *TRAPPC4*, and *PDXK*). Follow-up analysis examined genetic variation at each locus for association with PD, identifying a variant within the *PDXK* (pyridoxal kinase; rs2010795) gene conferring an increased risk of disease in independent patient-control series from Germany, Britain, and Italy. The combined series totaling 1,232 patients with PD and 2,802 control subjects resulted in $p = 1.18 \times 10^{-7}$ and odds ratio = 1.304 (95% confidence interval, 1.178–1.444).

As with all nominated genes for disease, independent replication is a prerequisite before pathogenicity can be confirmed. In addition, it is crucial to determine the global impact, prevalence, and population attributable risk. To this end, we examined the association of *PDXK* rs2010795 in 6 independent patient-control series from Europe, North America, and Asia (Supplemental Table). Genotyping was performed on the Sequenom platform, with a call rate of >95% for all series. Allele and genotype frequencies for each series and a combined analysis are displayed in Table 1.

We observed no significant association for rs2010795 and PD risk in our series. Although our findings do not rule out a role for other *PDXK* variants in susceptibility to PD, they do highlight the need for careful interpretation of genetic associations. The functional genomic screen employed by Elstner et al is an attractive approach; however, it is noteworthy that none of the known PD genes (*SNCA*, *LRRK2*, *PRKN*, *DJ-1,* and *PINK1*) was observed to display differential expression between the substantia nigra neurons of patients with PD and control subjects.

The recent advances in genetic technologies (eg, whole genome and exome sequencing[4]) may reduce the burden on the collection of expanded multi-incident pedigrees for the identification of the next PD gene.

TABLE: Allele and Genotype Frequencies of *PDXK* rs2010795

Series	Affection Status	Samples No.	Genotype			Allele		Odds Ratio (95% CI)	*p*
			GG	GA	AA	G	A (%)		
Taiwan	Patient	391	107	205	79	419	363 (46)	1.01 (0.82-1.24)	1.00
	Control	344	106	158	80	370	318 (46)		
USA[a]	Patient	314	152	128	34	432	196 (31)	1.04 (0.82-1.31)	0.76
	Control	348	173	138	37	484	212 (30)		
Canada[a]	Patient	527	259	220	48	738	316 (30)	1.03 (0.83-1.28)	0.84
	Control	318	158	138	22	454	182 (29)		
Ireland[a]	Patient	327	143	157	27	443	211 (32)	0.94 (0.75-1.18)	0.66
	Control	355	158	156	41	472	238 (34)		
Norway[a]	Patient	453	216	193	44	625	281 (31)	1.01 (0.84-1.23)	1.00
	Control	533	256	226	51	738	328 (31)		
Poland[a]	Patient	356	181	144	31	506	206 (29)	1.02 (0.81-1.28)	0.92
	Control	353	185	134	34	504	202 (29)		
Combined series	Patient	1,977	951	842	184	2744	1210 (31)	1.01 (0.91-1.11)	0.92
	Control	1,907	930	792	185	2652	1162 (30)		

Genotype and allele frequencies for *PDXK* rs2010795 in 6 independent populations. The ethical review boards at each institution approved the study, and all participants provided informed consent. Total sample sizes given for each series account for genotyping failure, which occurred in <5% of samples. Estimated odds ratios and *p* values were calculated under an allelic model. Meta-analysis with the results of Elstner et al showed significance ($p = 0.0002$); however, given our results this significance is driven entirely by the initial findings. The consistency in minor allele frequencies across our patient-control series does not support a role for this association in PD.
[a]Caucasian series used in the combined analysis.
CI = confidence interval.

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Additional Supporting Information may be found in the online verison of this article.

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The *PDXK* rs2010795 Variant Is Not Associated with Parkinson Disease in Italy

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We read with interest Elstner's article reporting the association of a polymorphism in the pyridoxal kinase (*PDXK*) gene with the risk of Parkinson disease (PD).[1] The authors used whole-genome expression profiling of dopaminergic neurons to identify 4 genes (*mtND2*, *SRGAP3*, *TRAPPC4*, and *PDXK)* differentially expressed between PD patients and controls. *PDXK* and *TRAPPC4* were further studied by association analysis in a German population (676/972 cases/controls) and in 2 replication cohorts, 1 British (203/1,126) and 1 Italian (353/704). A polymorphism (rs2010795) in *PDXK* was consistently associated with PD, reaching a combined p value of 1.2×10^{-7} (odds ratio [OR] = 1.3).

These exciting results awaited an independent replication; therefore, we decided to study the association of rs2010795 in an Italian PD cohort collected by the biobank of the Parkinson Institute of Milan (http://www.parkinson.it/dnabank.html). Our population included 920 patients (mean age, 63.5 $\pm$ 10.8 years; mean age of onset, 56.1 $\pm$ 11.0 years) and 920 controls (mean age, 62.5 $\pm$ 9.3 years), with no family history for movement disorders. Three hundred six patients were classified as familial, having at least 1 first-or second-degree relative with PD; the remaining were consecutively collected sporadic cases. Power estimates indicated that, if the analyzed polymorphism (minor allele frequency of ~30%, based on HapMap data on Caucasians) were to directly confer a 1.3-fold increase in the relative risk of PD (based on data from Elstner and colleagues[1]), the case and control groups used in this research would be of sufficient size to have 96% power to detect a significant association at the 0.05 level.

Genotyping (95% success rate) was performed by the high-resolution melting technique[2] using a LightCycler480 (Roche Applied Science, Indianapolis, IN). A 100% concordance was found after resequencing 10% of samples.

A standard case-control chi-square test was run to provide asymptotic p values and ORs for the minor allele (frequency, 33.14 and 33.20% for cases and controls, respectively). No evidence of association was found (p = 0.99; OR = 1; 95% confidence interval [CI], 0.93–1.07). Similar results were obtained analyzing familial (p = 0.97; OR = 1; 95% CI, 0.82–1.23) and sporadic (p = 0.94; OR = 0.99; 95% CI, 0.85–1.17) patients separately. Moreover, the distribution of genotype frequencies between cases and controls was not significantly different (p = 0.15).

Remarkably, both replication cohorts reported by Elstner included controls from isolated populations with a high level of endogamy, as thoughtfully acknowledged by the authors, a study design potentially prone to population stratification errors.

In this regard, our cohort is composed of patients originating from various Italian regions, and the controls are in most cases geographically matched.

Independent replication of new genetic associations is an essential step to confirm true-positive results and to better assess the conferred risk, which might be overestimated in the first study.[3] In our cohort, we found no evidence of association of PD with rs2010795 in *PDXK*. This is not sufficient to rule out a possible involvement of this gene in PD, but suggests cautiousness in considering rs2010795 a disease-associated marker, stressing the need for additional replications.

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